

# Edge4AV: Is Today's Edge Ready for Tomorrow's CAVs?

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## Abstract

Connected and autonomous vehicle (CAV) applications increasingly use information from nearby vehicles, roadside infrastructure, and edge services. It remains unclear whether today's edge communication technologies can satisfy the timing, response-availability, and payload requirements of tomorrow's CAV applications. We distinguish applications that consume independent periodic updates, including Basic Safety Message (BSM) and Signal Phase and Timing (SPaT) messages, from collaborative applications that require a response correlated with a particular request or session. The Intersection Programming Interface (IPI) represents both patterns through typed envelopes that record source, time, communication path, session, request correlation, payload, and response status. We evaluate these communication patterns using certified production C-V2X OBU/RSU devices, a Cisco private 5G deployment, and payloads derived from public CAV datasets. As normalized V2X signal strength decreased across stationary locations, the 2 KiB failure rate rose from 0.2% to 99.2%, while the 1 KiB failure rate rose from 0% to 2.1% and then 82.9%. Even 1 KiB delivery therefore required strong signal, while 256–512 B packets remained more reliable until the radio path became unavailable. On the 5G path, the compact workload group missed a 100 ms evaluation budget in 62.24% of attempts but missed a 500 ms budget in 3.71%; payloads of at least 1 MiB missed the 500 ms budget in every attempt. These results lead to three insights. 1) Current V2X technologies cannot support complex CAV applications. 2) 5G networks can support some complex CAV applications, but the data size is strictly limited. 3) Supporting collaborative CAV applications requires new networking or communication methods beyond using today's V2X and 5G network paths as-is.

## Keywords

connected and autonomous vehicles, V2X, private 5G, edge computing, smart intersections

## 1 Introduction

Connected and autonomous vehicles (CAVs) can use information beyond the field of view of their onboard sensors. At an intersection, this information can include traffic-signal timing, the motion of nearby vehicles, and warnings about road users occluded by buildings, stopped traffic, or other vehicles. Recent work on smart intersections shows how roadside infrastructure can provide such information to automated

vehicles [15, 22, 34, 35]. The value of this infrastructure information depends on communication: an application cannot act on information that is missing, stale, or delivered after its decision time.

Current V2X standards support periodic vehicular communication. The Basic Safety Message (BSM) reports vehicle state, while the Signal Phase and Timing (SPaT) message reports traffic-signal state and timing [3, 32, 43]. An FHWA spectrum study uses a typical size of 320 B for both messages, and connected-intersection guidance recommends keeping SPaT and MAP messages small to maximize reception reliability [10, 16]. Applications that consume these messages follow a stateless communication pattern. The receiver acts on the newest valid update, and a new update supersedes an older one without completing an exchange associated with the older message. Many proposed collaborative CAV applications use a different pattern. Systems for infrastructure-assisted perception, perception fusion, real-time mapping, and planning support exchange application data while a driving task is in progress [20, 21, 23, 46, 47, 56].

Many of these applications require a response associated with a particular request or session. We call applications with this dependency stateful. A stateful application may request data for a road region, wait for detector output from infrastructure, combine responses from multiple participants, or use a response to select its next action. Its communication requirement includes response availability, a time budget, and the size of the application data, not only delivery of the newest periodic update. Here, *complex CAV applications* denotes these stateful collaborative applications.

The two communication patterns require different measurements. Periodic updates are evaluated by whether sufficiently fresh updates continue to arrive. A stateful exchange also requires the response to a specific request to arrive before the application's time budget expires. Measuring only the latency of successful packets can therefore hide missing responses and deadline misses. The resulting question is: *can today's edge communication technologies support tomorrow's CAV applications across both communication patterns?*

To answer this question, we measure a certified production C-V2X OBU/RSU path and a Cisco 5G network path to an edge service. C-V2X provides direct vehicle-to-infrastructure communication, while 5G provides an addressed IP path between a vehicle and an edge endpoint [6, 8, 13, 36, 44, 48]. The V2X experiments measure compact payloads under different stationary propagation conditions and along mobile routes. The 5G experiments measure TCP, MQTT over TCP,

and UDP across payload sizes, background load, signal conditions, detector-output replay, client counts, service restarts, and evaluation time budgets. Public CAV datasets and a detector benchmark provide the application-object sizes used to interpret and select 5G payload conditions [12, 33, 42, 49, 51, 53, 55].

The V2X results show a joint dependence on packet size and signal strength. As normalized signal strength decreased across the stationary points, the 2 KiB failure rate increased from 0.2% to 99.2% and then reached 910 failures in 910 observed attempts. The 1 KiB failure rate increased from 0% to 2.1% and then 82.9%, showing that even 1 KiB delivery requires a strong signal. An intervening building makes Point 1 non-line-of-sight (NLOS), which explains its weak signal despite its short distance from the RSU. On the 5G path, response time increased with payload size, load, weak signal strength, and concurrency. Dataset-derived detector, image, point-cloud, and map objects exceed the scale of compact vehicular messages, and the latency of one transfer chunk does not establish complete-object latency.

These results lead to three insights. 1) Current V2X technologies cannot support complex CAV applications. 2) 5G networks can support some complex CAV applications, but the data size is strictly limited by the application's time budget and operating conditions. 3) Supporting collaborative CAV applications requires new networking or communication methods beyond using today's V2X and 5G network paths as-is. Such methods can include deadline-aware path selection, payload reduction, caching or prefetching, and transfer scheduling rather than sending every application object unchanged on a single path.

The contributions are:

- We introduce IPI, a typed, transport-independent interface for periodic V2X updates and correlated CAV service exchanges. Its envelope, session, correlation, freshness, payload, and status fields, together with explicit failure semantics, provide a common contract for both communication patterns.
- We characterize certified production V2X and Cisco 5G paths across radio condition, payload size, protocol, load, concurrency, service restarts, and evaluation time budgets, and relate the measurements to dataset-derived CAV workloads.

## 2 Related Work

Related work spans V2X standards and interfaces, V2X radio measurement, 5G and edge networking, and infrastructure-assisted CAV systems. Across these areas, the unresolved issues concern deployed-system behavior, response loss, effects of operating conditions, and transfer of application objects (Table 1).

**V2X standards and interfaces.** SAE J2735 defines message content such as BSM, SPaT, MAP, SRM, and SSM, while 3GPP specifications and surveys describe V2X service requirements, architectures, and radio evolution [1–5, 7, 18, 19, 29–32, 40, 41, 43, 45]. These standards establish interoperable message and service semantics. Their deployed message scale is compact: FHWA reports typical BSM and SPaT sizes of 320 B, and CTI 4501 permits a C-V2X payload of up to 8000 B including security overhead while recommending small SPaT and MAP messages to maximize reception reliability [10, 16]. Device interfaces impose implementation-specific limits. For example, the Ettifos V2X-AIR manual reports an 8188 B non-IP MTU and a 1500 B IP MTU [14]. These values differ across implementations, but all of the reported limits require application objects larger than 8 KiB to be segmented or carried over another network path. IPI preserves standard V2X payloads and adds transport-independent envelope, session, request, and response operations with explicit freshness, correlation, payload, status, and failure semantics.

**V2X radio measurement.** Prior measurement studies characterize vehicular links under particular physical-layer and mobility conditions; millimeter-wave V2X work, for example, measures connectivity under directional links and motion [50]. Application-aware testing work also shows that communication behavior should be evaluated together with the application action it enables [17]. Complementing these studies, measurements on certified production C-V2X equipment distinguish the RTT of completed request/response probes from response availability across all attempts. This distinction is necessary when completed probes remain fast but many requests receive no response.

**5G and edge networking.** Prior work studies 5G-V2X, network slicing, MEC-based vehicular computing, wide-area cellular performance, cloud-core behavior, and latency control for edge analytics [6, 8, 9, 13, 34–36, 44, 48]. These systems establish that cellular and edge platforms can host useful CAV services. The vehicle-to-edge measurements characterize a different boundary: response time and response availability as payload size, TCP/MQTT/UDP protocol condition, signal strength, background load, client count, and service restarts change.

**Infrastructure-assisted CAV systems.** Infrastructure-assisted systems determine the application objects represented in the workloads. Prior systems explore edge-assisted perception, multi-agent perception fusion, smart roadside infrastructure, real-time maps, planning support, and maneuver generation [11, 20, 21, 23–25, 28, 33, 37–39, 46, 47, 51–56]. Public CAV datasets and detector output quantify the data sizes implied by these application classes. The resulting evidence characterizes communication paths and maps workload sizes; it does not propose a new perception, mapping, planning, or scheduling system.

**Table 1: Prior work and remaining CAV communication measurement issues.**

| Research area                                                       | Established contribution                                                                     | Remaining measurement issue                                                                                                                                    | Measured evidence                                                                                                                   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| V2X standards and interfaces<br>[3, 6, 32, 41, 43]                  | Message formats, service requirements, radio-access evolution, and deployment architectures. | Whether a current deployed path can satisfy request-specific timing, response-availability, and payload requirements in addition to periodic message delivery. | Standard BSM/SPaT payload preservation; typed IPI update, session, request, and response contracts with explicit failure semantics. |
| V2X radio measurement<br>[17, 50]                                   | Link behavior under controlled physical conditions and application-aware testing methods.    | How latency conditional on receiving a response relates to the fraction of requests that receive no response.                                                  | Separate successful-response RTT and attempt-level response availability under stationary and route-based signal conditions.        |
| 5G and edge networking<br>[8, 13, 34–36]                            | MEC architecture, vehicular edge services, network slicing, and edge analytics.              | How request/response performance changes with CAV payload size, protocol, radio condition, offered load, concurrency, and service interruption.                | Response time and availability across these factors on a Cisco deployment, with 100, 500, and 1000 ms time budgets.                 |
| Infrastructure-assisted CAV systems<br>[20, 21, 23, 37, 46, 47, 56] | Cooperative perception, roadside intelligence, mapping, planning, and collaborative driving. | Whether the communication paths assumed by these systems can transfer the required application objects within their decision time.                             | Dataset and detector object-size mapping to measured 5G payload conditions.                                                         |

### 3 Intersection Programming Interface

Interpreting a network trace against application requirements requires more than packet timestamps and sizes. The trace must identify whether a packet is an independent update or part of a request that expects a correlated response; it must also preserve the application payload, freshness information, response state, and session context. IPI provides this contract as a typed, transport-independent C++17 interface. A common `Envelope<T>` identifies the message, source, intersection, send time, and communication path, while the enclosed type defines a stateless V2X update or a stateful service operation. IPI is not a perception system or planner; it defines how those applications identify and exchange application messages.

The reference API exposes six operations: ingest a typed J2735 envelope, register or terminate a session, send a heartbeat, invoke a service, and retrieve correlated responses. Service-request and cooperative-service types have canonical encoders, decoders, and validators. Contract fields are classified as required after envelope normalization or as optional application context. The implementation generates `messageId` and `sentAt` when the caller omits them.

**State model and failure semantics.** A stateless update has no request lifecycle: the receiver accepts or discards the update according to message validity and freshness, and the next valid update supersedes it. A stateful exchange creates a pending request, optionally within an active session. A session-scoped exchange enters `ACTIVE` at registration and remains active through heartbeat, patch, and telemetry operations. Responses may report `ACCEPTED` or `IN_PROGRESS` before reaching `COMPLETED` or `REJECTED`. The type system

can also represent a suspended session, although that state is not exercised in the measurements. A response that expires is stale, whereas a sender timeout means that no correlated response arrived before the local wait ended. A timeout, packet loss, or service restart therefore cannot be relabeled as an explicit rejection.

**Concrete request/response example.** A perception-aid request can use request 17 for vehicle `veh-9` in session `s-9`. It carries the vehicle location, an optional 2500 ms planning horizon, and a region query in the application payload. Its envelope supplies the source, intersection, path, and send time and uses `correlationId=17`. A corresponding response retains the session and correlation identifiers and carries `status=COMPLETED`, an optional `expiresAt`, and, for example, a 19.2 KiB detector-object list. The response can therefore be matched to the request even when unrelated periodic updates and other service exchanges use the same communication path.

**Analysis enabled by IPI.** An ordinary IP packet log records endpoints, protocol, timestamps, and bytes, but it does not by itself identify application sessions, correlation between requests and responses, response status, or freshness. IPI preserves those semantics in typed objects and exposes their identifiers to the experiment log. Matching `correlationId` yields sender-side RTT, while retaining every `requestId` supplies the denominator for response availability and time-budget misses. Response status and the optional expiration time distinguish explicit rejection or stale data from a missing response; run and condition identifiers

### IPI API contract, lifecycle, and recorded measurement fields.

| Object and API                                     | Required fields                                                                                       | Optional fields                                                                                                                | State or failure semantics                                                                                                                                   | Recorded measurement fields                                                                                       |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| StatelessUpdate;<br>ingestV2xMessage               | messageId, sentAt, intersectionId, transport, source{type, id}, and J2735 type, encoding, and payload | sessionId, correlationId, priority, frameCounter, and radio diagnostics                                                        | Independent update; a newer valid update supersedes an older one. Local ingestion returns Ack.                                                               | BSM/SPaT validation, V2X compact-payload baseline, and 5G SPaT mirror                                             |
| SessionRegister,<br>heartbeat,<br>terminateSession | Registration envelope and vehicle id/role; sessionId for subsequent operations                        | VIN, OEM/version, requested services, fallback path, RSSI threshold, subscription, preferred channels, and heartbeat telemetry | Register → ACTIVE; heartbeat/patch retain ACTIVE; terminate → TERMINATED.                                                                                    | Session identifier in 5G logs; registration and heartbeat latency are not measured                                |
| ServiceRequest;<br>invokeService                   | serviceType, requestId, vehicleId, location, and common envelope fields                               | sessionId, VIN, speed, heading, desiredHorizonMs, application data, and priority                                               | Invocation creates a pending request. If absent, correlationId is generated from requestId. The planning horizon is not a network deadline.                  | Request/sequence identifier, payload bytes, service, protocol condition, and 100/500/1000 ms comparison threshold |
| ServiceResponse;<br>listSessionResponses           | correlationId, serviceType, vehicleId, and status                                                     | expiresAt and guidance or result payload                                                                                       | ACCEPTED/IN_PROGRESS are nonterminal; COMPLETED/REJECTED are terminal. Expiration marks a response stale.                                                    | Correlated response, status, response bytes, and sender-side RTT                                                  |
| Ack; timeout, missing response, and restart        | Local accepted value and sender attempt identifier                                                    | Identifier or validation/error text                                                                                            | Local validation failure and remote REJECTED status are distinct. Timeout, missing response, and restart are communication outcomes, not rejection statuses. | Response-availability denominator, time-budget miss, and restart outcome; no explicit service-rejection trials    |

separately identify restart-affected attempts. The same analysis can therefore be applied to V2X and 5G traces without treating every received packet as an interchangeable success.

One reference composition attaches a service request to a vehicular update, maintains session context at the infrastructure, and returns a correlated edge response (Figure 1). The measured system boundary consists of the direct OBU-RSU V2X path and the vehicle-edge 5G path, evaluated independently rather than as the complete workflow.

V2X validation records J2735 update types, while compact probes retain request and sequence identifiers, payload size, timeout outcome, and RTT. The 5G records add session and service identifiers, protocol condition, response status, sender timestamps, and payload bytes. Detector and dataset-derived replay retain the same request, payload, response, and timeout fields. Session registration and heartbeat are implemented; their setup latency and the complete workflow in Figure 1 are outside the measured boundary.

## 4 Testbed Design

Two communication paths are measured independently in the same road environment. The first is a direct C-V2X path between an OBU and an RSU. The second is a vehicle-to-edge path through a 5G deployment comprising a Cisco edge mobile core, an Airspan AirSpeed 2900 n48 gNodeB, and a Cisco Meraki MG52-HW vehicle gateway. Public CAV datasets and detector output supply payload sizes; they are workload inputs rather than a third network path. No request traverses one path and returns over the other.

**Table 2: Measured communication paths and workload inputs.**

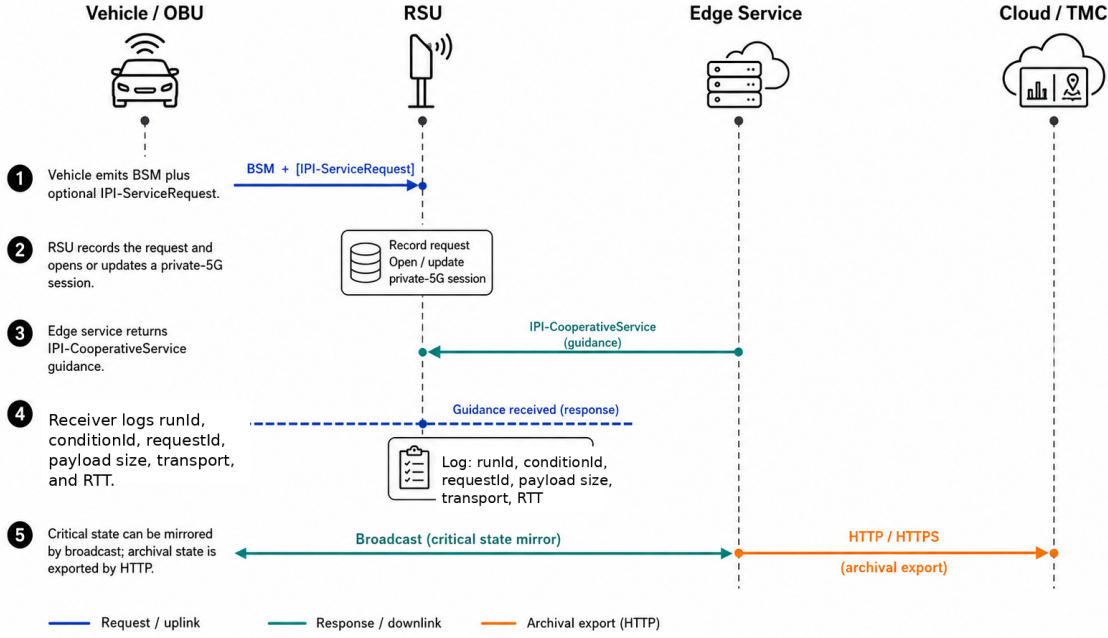
| Component            | Function                                                                                                                                                                           |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certified C-V2X path | Direct OBU-RSU communication. Standard BSM/SPaT operation validates the production radio path; a custom packet-data request/response probe measures RTT and response availability. |
| 5G path              | Vehicle-side sender through the gateway, radio, and core network to a fixed edge-side host running TCP, MQTT-over-TCP, or UDP services.                                            |
| Workload inputs      | Detector output and public CAV artifacts that determine the 5G application payload sizes.                                                                                          |

### 4.1 Measurement Environment

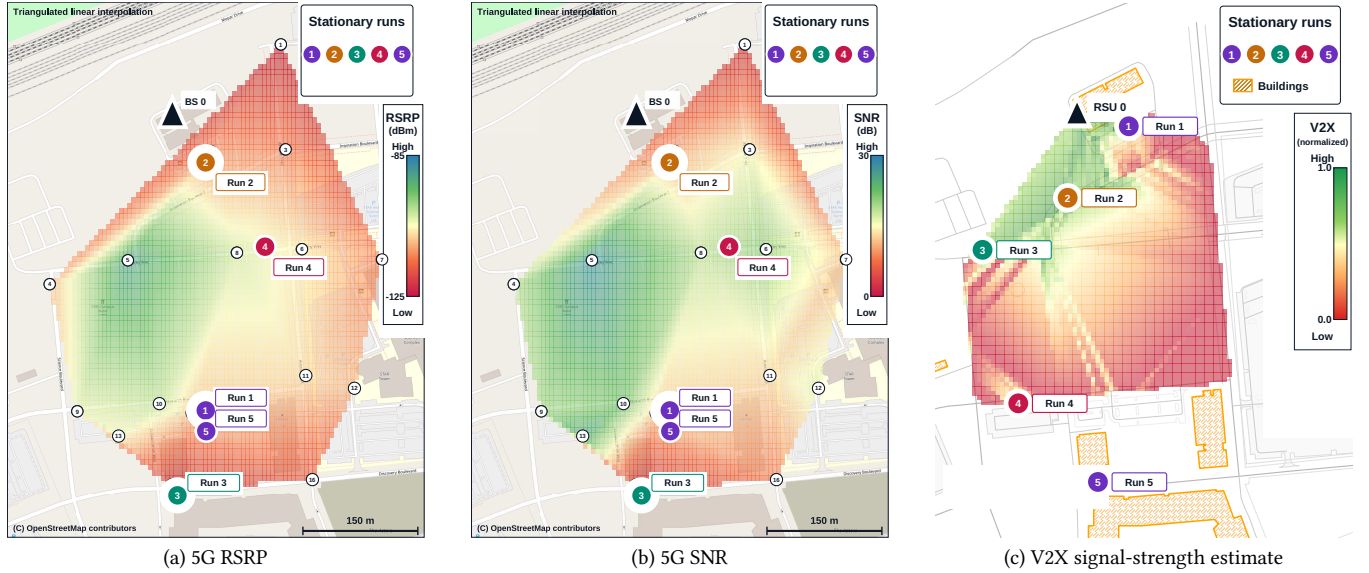
Wireless measurements were collected around fixed roadside infrastructure. The area contains a fixed V2X RSU, a fixed 5G base station, stationary collection points, mobile routes, and nearby buildings. The direct path between the RSU and Point 1 intersects a building footprint, making Point 1 NLOS and severely attenuating its signal despite the short distance. Figure 2 depicts the spatial radio context using anonymized coordinates.

### 4.2 V2X Radio Path

The V2X path uses certified production Mocar C-V2X devices: an OBU on the vehicle and a fixed RSU. The vendor identifies Mocar as its V2X terminal and SDK product family



**Figure 1: Reference IPI workflow for a stateful intersection service. The workflow composes update, request, session, response, broadcast, and archival operations. Measured results correspond to the isolated V2X and 5G communication paths rather than the complete end-to-end workflow.**



**Figure 2: Radio conditions in the measurement area. Panels (a) and (b) interpolate 5G RSRP and SNR. Panel (c) shows normalized V2X signal strength. Numbered circles identify stationary V2X locations; orange hatching marks buildings that obstruct LOS paths.**

and reports LTE-V2X protocol-conformance certification for the MOCAR OBU and HUALI RSU [26, 27]. Standard-path validation delivered 10/10 SPaT frames to the OBU callback

and verified bidirectional BSM reception. The latency experiments use the same radio stack with a vendor packet-data responder so that the sender can associate each reply with

a request and measure RTT. Static analysis of the packet-data SDK installed on both endpoints identified a maximum application packet of 4,080 B after vendor framing. This is an implementation limit rather than a C-V2X standard limit. The probe is not a BSM or SPaT message; it measures the direct path used by the production devices.

Stationary conditions send 1000 requests per payload at a 100 ms interval, vary payload from 0 to 2048 B, and use a 1000 ms timeout at locations spanning strong to weak normalized signal. Mobile conditions use a 256 B payload at a 200 ms interval and a 500 or 1000 ms timeout; two runs had median speeds of 7.13 and 7.54 m/s. Their GNSS-joined latency and failure samples produce the normalized V2X signal-strength map in Figure 2(c).

### 4.3 5G Network Path

The 5G path starts at a vehicle-side sender connected through the MG52-HW, traverses the Airspan gNodeB and Cisco core, and terminates at a fixed edge-side host on the 5G subnet. The MG52-HW remained flat on the vehicle's trunk floor, with its front face upward, throughout the follow-up measurements. The edge host runs the TCP and UDP responders; for MQTT, it also runs the MQTT 3.1.1 QoS 0 broker and receiver. No cloud service or Internet backhaul is included in the measured RTT. Sender timing begins after request encoding and immediately before transmission, and it ends when the correlated application response is received.

Most 5G conditions contain 1000 attempts at a 200 ms interval. Payload sweeps span 0 B to 2 MiB. Multiclient conditions use 1–100 clients, each offering a nominal five requests per second, for an aggregate offered rate of 5–500 requests per second. Configured sender timeouts vary by protocol and experiment from 3 to 60 s. These harness timeouts can exceed the 100, 500, and 1000 ms analysis budgets defined in Section 5. Endpoint clocks were generally unsynchronized, so sender-side RTT is the latency metric rather than one-way delay.

The original experiments used the 40/40/20 TDD profile. The follow-up used the operator-reported 70/20/10 profile with the MG52-HW locked to Airspan Cell 2 and the same 10D4G setting. Unaligned management-plane exports preclude a controlled causal comparison between the profiles.

### 4.4 Workload Inputs

The workload inputs include detector result objects, encoded images, point clouds, and HD-map files. Detector-output percentiles are replayed directly as 5G request payloads. For larger image, point-cloud, and map objects, fixed 128–512 KiB and 1–2 MiB conditions measure one transfer size or chunk; they do not measure completion time for every full source

object. The workload mapping therefore treats these conditions as transfer sizes or chunks rather than complete-object latency.

## 5 Experiment Design

Readiness is characterized along four measurable dimensions: response availability, sender-side RTT, sensitivity to payload and operating condition, and attainment of fixed time budgets. Five experiment groups vary the communication path, payload, protocol, radio condition, load, concurrency, and service availability (Table 3).

### 5.1 Metrics and Analysis Rules

*Response availability* includes every sender attempt. It equals the number of attempts that receive a correlated application response before the configured sender timeout divided by the total number of attempts. Sender-side RTT is defined only for attempts that receive a response. Separate response-availability and RTT metrics prevent low RTT among completed exchanges from concealing requests that produced no response.

The 100, 500, and 1000 ms thresholds are common evaluation budgets, not application-specific standards or guarantees. For a given budget, an attempt is a hit only if it receives a response and its RTT does not exceed the budget. A timeout, missing response, or response later than the budget is a miss. The configured sender timeout controls how long the harness waits and can be much longer than an evaluation budget. Consequently, an attempt can count as available but still miss a time budget.

Most 5G sender and receiver clocks were not synchronized. Sender-side request/response RTT therefore provides the timing metric, while one-way fields are retained only as diagnostics. GNSS-joined latency and packet failures from the mobile V2X runs form a normalized signal-strength map. The stationary payload sweeps are separate runs evaluated at locations on this map.

### 5.2 V2X Radio Experiments

The V2X setup first validates standard operation by receiving SPaT frames and BSMs through the production device callbacks. The latency experiments then use a custom packet-data request/response probe on the same certified OBU/RSU radio path. The custom probe supplies sequence and request identifiers needed to associate each reply with its transmission; it is not labeled as a standard BSM or SPaT message.

Stationary runs vary request payload from 0 to 2048 B. Each condition contains 1000 attempts at a 100 ms interval with a 1000 ms timeout. The locations span strong to weak normalized signal, including building-obstructed Point 1. The 256 and 512 B conditions represent compact vehicular

**Table 3: Experiment groups and measured factors.**

| Experiment group                  | Factors varied                                                                          | Metrics                                                                               | Communication property                                                                      |
|-----------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Stationary V2X probes             | Payloads from 0 to 2048 B at fixed LOS and NLOS locations                               | Response availability and successful-response RTT                                     | Payload- and propagation-condition dependence of compact direct-radio exchange              |
| Mobile V2X probes                 | 256 B probes along routes containing available, weak, and unavailable link segments     | Route-level response availability, successful-response RTT, and spatial link estimate | Route-level availability of compact direct-radio exchange                                   |
| 5G payload/protocol sweeps        | TCP, MQTT over TCP, and UDP; payloads from 0 B to 2 MiB                                 | Response availability, RTT percentiles, and time-budget miss rate                     | Protocol- and payload-dependent addressed vehicle-to-edge exchange                          |
| 5G stress and restart experiments | Background uplink load, signal strength, 1–100 clients, and receiver or broker restart  | Tail RTT, response availability, and time-budget miss rate                            | Compact-exchange stability under load, weaker signal, concurrency, and service interruption |
| Dataset-derived 5G replay         | Detector-output percentiles and fixed transfer sizes selected from public CAV artifacts | Response availability and RTT for the replayed size; source-object size distribution  | Correspondence between CAV application objects and measured payload conditions              |

messages; the 1 and 2 KiB conditions determine whether larger packets require stronger signal. Mobile runs use a 256 B payload at a 200 ms interval and a 500 or 1000 ms timeout to sample signal strength along the route.

### 5.3 5G Network Experiments

The 5G experiments use an addressed IP path from the vehicle-side sender to the edge host. TCP uses a receiver that returns one response per request. MQTT uses a QoS 0 broker and application receiver over TCP. UDP uses either one datagram per message or fragmentation and reassembly in the experiment harness. These are *protocol conditions*; MQTT is not a transport protocol independent of TCP.

Payload sweeps begin with 0–4 KiB compact exchanges and extend through detector-output sizes, 128–512 KiB transfer sizes, and 1–2 MiB bulk sizes. Each payload size contributes response availability and RTT percentiles, and all attempts are then evaluated against the three time budgets. This separates an eventual response from a response delivered within a specified time.

Stress experiments keep the payload at 1 KiB while varying operating condition. Background-load runs add uplink traffic. Signal experiments repeat the same service traffic under lower measured RSRP and SNR. Multiclient runs use 1–100 clients; each client sends 1000 requests at a nominal 200 ms interval, producing an aggregate offered rate from 5 to 500 requests per second. Service-interruption runs restart the TCP or UDP receiver or the MQTT broker during active probes. They measure restart loss and recovery; no fallback path is exercised. QoS-label runs record the requested label

and separately inspect host packet markings, so the label alone does not establish network-enforced QoS.

The locked-cell follow-up repeats four TCP, MQTT, and UDP conditions: C1 is an idle 1 KiB exchange, C2 uses the 23,968 B detector-output p99, C3 adds an offered 25 Mbps vehicle-to-edge load to C1, and C4 uses 100 clients sending 500 1 KiB requests each. Each repetition is a separate five-minute interval. C1 and C2 are repeated at three stationary locations with fixed gateway placement and network configuration. Application RTT and availability come from validated sender logs; location classes remain operator-reported pending matched exports.

### 5.4 Dataset-Derived 5G Workloads

The workload manifest characterizes annotations, encoded images, point clouds, and HD-map objects from the staged CAV datasets. A detector benchmark adds a structured result whose estimated serialization contains a 256-byte header and 96 bytes per predicted object. The observed minimum, p50, p95, p99, and maximum detector-result sizes are replayed directly over the 5G path, together with a 60 KiB upper condition.

Many images, point clouds, and HD-map objects exceed one IPI service payload. The 128–512 KiB and 1–2 MiB sweeps therefore measure fixed transfer sizes or chunks selected from their size distributions. They do not measure the time to deliver every complete multi-megabyte object. Detector execution time is measured separately and excluded from network RTT. The detector replay thus evaluates post-inference

communication, not the latency of the complete sensing, inference, and transfer pipeline.

## 5.5 Communication Requirements

The V2X conditions characterize compact direct-radio exchange under the observed signal conditions. The 5G conditions characterize addressed vehicle-to-edge exchange across payload, protocol, load, signal strength, concurrency, interruption, and time budget. Workload sizes connect these network conditions to detector, image, point-cloud, and map data. Their combination tests both periodic compact-update requirements and request-specific delivery of application data.

## 6 Experiment Results

RTT percentiles characterize attempts that receive a response. Response availability and time-budget miss rates include every sender attempt, thereby representing both completed-response timing and missing responses.

Figure 3 spans fixed and route-based V2X, compact and bulk 5G payloads, idle and stressed 5G conditions, multiclient load, and service restarts. Table 4 summarizes the repeated locked-cell 5G follow-up.

### 6.1 V2X Radio Results

The stationary and mobile V2X experiments show that signal strength and packet size jointly limit delivery. Under the strongest stationary signal condition, the 256, 512, 1024, and 2048 B packets received 1000, 1000, 1000, and 998 responses from 1000 attempts. Median RTT was approximately 100 ms at every size.

As normalized signal strength decreased across Points 2–4, the 2 KiB failure rate rose from 0.2% to 99.2% and then reached 910 failures in 910 observed attempts. For 1 KiB packets, failure rose from 0% to 2.1% and then 82.9%. The corresponding 512 B failure rates were 0%, 0.2%, and 3.1%, and the 256 B rates were 0%, 0%, and 0.9%. At Point 5, even the minimum packet condition received no responses. Point 1 also had low normalized signal because a building blocked the direct LOS path; none of its 900 observed 1 KiB attempts completed before the run was stopped. Thus, lower signal strength reduces the packet size that V2X can deliver reliably, and even 1 KiB packets require a strong signal.

Across the four route runs, Run 1 ended after the first drive and therefore contains 675 attempts; the other runs contain 1000. Runs 2–4 show that completed responses can retain 28–29 ms median RTT even while 26.9–40.9% of requests receive no response.

#### V2X mobile-route results across all four runs.

| Run | Attempts | Resp. avail. | p50 / p95 RTT (ms) |
|-----|----------|--------------|--------------------|
| 1   | 675      | 76.3%        | 99.6 / 108.4       |
| 2   | 1000     | 73.1%        | 28.3 / 41.9        |
| 3   | 1000     | 59.1%        | 29.2 / 43.9        |
| 4   | 1000     | 71.6%        | 28.3 / 41.6        |

Standard-path checks connect the probes to deployed V2X operation. The certified production OBU/RSU pair delivered 10/10 standard SPaT frames to the OBU receive callback and received BSMs in both directions. Short custom packet-data diagnostics also returned 20/20 baseline replies and 30/30 replies in each diagnostic mode, with p95 RTT of approximately 50–58 ms. The installed SDK’s public custom-send symbol did not transmit in this build, so the RTT probe used the vendor packet-data function used by the working radio stack. This implementation detail changes the probe entry point, not the measured physical OBU–RSU path.

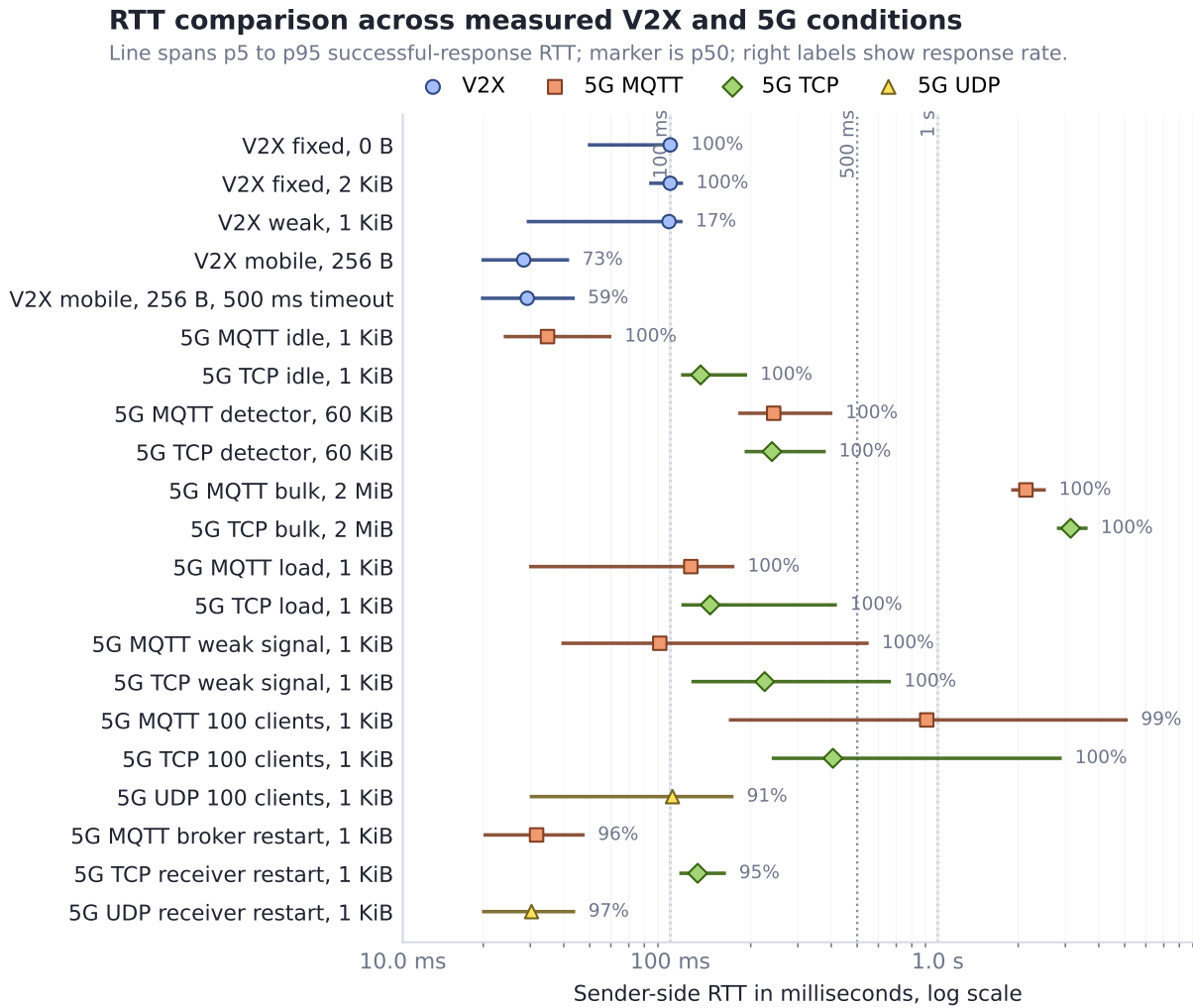
Typical BSM and SPaT messages are approximately 320 B [16], matching the 256–512 B conditions that remained usable under weaker signal. The smallest detector output in the workload set is 13,216 B, and images, point clouds, and maps are larger still. These objects require multiple packet deliveries, while the measurements show that even one 1 KiB packet requires strong signal for reliable delivery. Current V2X technologies can carry compact periodic messages but cannot support complex CAV applications.

### 6.2 5G Payload and Time-Budget Results

The 5G payload experiments measure an addressed vehicle-to-edge path. For an idle 1 KiB MQTT exchange, p50 RTT was 34.8 ms and p95 RTT was 60.2 ms. With 25 Mbps of background uplink traffic, the same payload had p50 RTT of 119.4 ms and p95 RTT of 173.8 ms. A 60 KiB detector-output-sized MQTT exchange had p50 RTT of 243.7 ms and p95 RTT of 403.8 ms. The 2 MiB MQTT and TCP conditions had seconds-scale RTT.

Stationary repeats show that payload size and radio condition interact. Under the stronger measured signal conditions, 0–4 KiB MQTT p95 RTT was approximately 36–49 ms and TCP p95 RTT was approximately 100–155 ms. In a weaker and more variable stationary condition, 0 B p95 RTT exceeded 500 ms for MQTT and one second for TCP. Large transfers remained seconds-scale across repeats; one weak-signal 256 KiB MQTT condition had p95 RTT above 40 s. Payload size sets the amount of data to transfer, while signal strength, load, and queueing determine the observed variation for that size.

All sender attempts are aggregated within each workload group (Table 5). The aggregation is attempt-weighted; it does not average condition medians. Multiclient attempts are excluded from the compact service row and form the separate aggregation in Table 7. Response availability uses



**Figure 3: V2X and 5G sender-side RTT across the measured conditions. Intervals span p5–p95 RTT among completed responses, markers denote p50 RTT, and right-hand labels give response availability across all attempts.**

**Table 4: Locked-cell 5G follow-up. Protocol cells report p50/p95 RTT in milliseconds and response availability. Common/typical pools two blocks; other rows contain two repetitions. Location classes and the Cell 2 lock are operator-reported pending matched exports.**

| Location class   | Workload                           | N/proto. | TCP              | MQTT             | UDP                |
|------------------|------------------------------------|----------|------------------|------------------|--------------------|
| Common/typical   | C1: 1 KiB, one client, idle        | 2000     | 123.7/149.9/100% | 30.4/41.6/100%   | 29.7/40.3/100%     |
| Common/typical   | C2: 23,968 B, one client, idle     | 2000     | 85.5/107.0/100%  | 59.8/75.8/100%   | 56.6/70.5/99.95%   |
| Weak candidate   | C1: 1 KiB, one client, idle        | 1000     | 126.8/150.4/100% | 32.5/46.1/100%   | 31.7/43.9/100%     |
| Weak candidate   | C2: 23,968 B, one client, idle     | 1000     | 152.5/188.7/100% | 126.0/162.9/100% | 120.6/148.6/99.40% |
| Strong candidate | C1: 1 KiB, one client, idle        | 1000     | 117.9/137.3/100% | 28.0/40.0/100%   | 26.6/39.2/100%     |
| Strong candidate | C2: 23,968 B, one client, idle     | 1000     | 75.4/93.6/100%   | 49.8/59.7/100%   | 47.9/56.9/100%     |
| Medium/typical   | C3: 1 KiB with offered uplink load | 10000    | 330.9/399.7/100% | 139.7/169.6/100% | 139.7/171.5/99.50% |
| Medium/typical   | C4: 1 KiB, 100 clients             | 100000   | 250.0/355.9/100% | 158.7/473.8/100% | 103.7/170.4/98.67% |

every attempt, and a missing response is a miss at all three evaluation budgets.

The compact-service group received responses for every attempt, but 62.24% of attempts exceeded the 100 ms budget. At 500 ms, its miss rate fell to 3.71%. The 128–512 KiB group

had a median RTT of 227.2 ms and missed the 100 ms budget in 99.01% of attempts. Payloads of at least 1 MiB had median RTT above one second and missed the 500 ms budget in every attempt. These thresholds are comparison points; the results do not assign one universal deadline to every application in a workload group.

Application objects are assigned to these groups by source size and replayed representation (Table 6). The same three budgets are applied to every row to expose sensitivity to the selected threshold; they are not source-dataset deadlines.

Detector-output percentiles were replayed directly. Under stronger signal, TCP and MQTT completed the 23–25 KiB conditions with p95 RTT below 214 ms. They also completed 60 KiB transfers, whose p95 RTT was approximately 382–404 ms. The unfragmented UDP implementation returned no replies at 1400, 4096, or 19648 B. The failure begins near the datagram path’s size limit, and adding fragmentation changes the outcome. It therefore identifies a limitation of the measured implementation and path, not a general property of UDP or 5G.

With application-level fragmentation and reassembly, UDP completed transfers up to approximately 24 KiB. At 60 KiB, it completed 20/100 attempts under the stronger signal condition and 0/100 under the weak-signal condition. Thus, the UDP result depends on representation, fragmentation, and radio condition. A datagram response with lower RTT is not useful if the complete response is not reassembled before the timeout or evaluation budget.

Weak-signal detector replay separates response availability from timeliness for the TCP-based conditions. TCP and MQTT returned responses for every detector-size attempt, including 60 KiB, but the 60 KiB p95 RTT increased to 662 ms for TCP and 770 ms for MQTT; p99 RTT reached 1027 and 1230 ms. Thus, complete response availability does not imply attainment of the 500 or 1000 ms budgets in the tails. Fragmented UDP showed the opposite tradeoff: 19–25 KiB availability fell to 93.5–96.1%, and the 60 KiB condition returned 0/100 responses.

The locked-cell follow-up adds 60 validated samples (Table 4). Its C1–C4 block received 307,662 of 309,000 responses. C3 p95 RTT was 169.6–399.7 ms under offered uplink load, with five UDP timeouts. At 100 clients, TCP and MQTT received all 100,000 responses per protocol, while UDP received 98,667; their respective 100 ms hit rates were 0.004%, 6.819%, and 47.102%.

For the C2 detector payload, candidate-weak p95 RTT was 148.6–188.7 ms and candidate-strong p95 RTT was 56.9–93.6 ms. Candidate-strong p95 was 2.84–21.16% below the pooled common/typical values across all six C1–C2 protocol aggregates. These sequential application-layer results do not establish an RSRP, load, or TDD effect without matched

exports. The R1 controls measured 13.118 Mbps vehicle-to-edge and 25 Mbps edge-to-vehicle host throughput.

### 6.3 5G Stress and Service-Restart Results

The 1 KiB stress experiments isolate operating condition from payload growth. With 25 Mbps of background uplink traffic, MQTT p95 RTT increased from 60.2 to 173.8 ms and TCP p95 RTT increased from 193.8 to 419.8 ms. Under the weak-signal idle condition, p95 RTT was 551.8 ms for MQTT and 668.3 ms for TCP. The 5G signal survey covered RSRP from -121 to -91 dBm and SNR from 3.5 to 27.5 dB, confirming that the collection locations included materially different radio conditions.

The load-stream sweeps did not produce a monotonic relationship between stream count and tail RTT, which prevents attributing the result to stream count alone. With 1, 2, and 4 default uplink streams under stronger signal, MQTT p95 RTT was 170, 468, and 182 ms, while TCP p95 RTT was 391, 402, and 410 ms. Repeating the same stream counts with the application-level 5QI label produced MQTT p95 RTT of 377, 487, and 529 ms and TCP p95 RTT of 431, 680, and 772 ms. A second weak-signal run had lower idle tails than the first, but loaded p95 RTT remained 460–708 ms for MQTT and 630–759 ms for TCP. These repeats show that background traffic and signal condition increase the range of observed tail RTT, while the stored data do not support a deterministic per-stream scaling law.

The QoS-label experiment produced a negative result. The harness recorded an application-level 5QI-mapped label, but packet captures decoded the observed packets with tos 0x0. The experiment therefore does not establish host-side marking or network-enforced 5QI treatment. The labeled conditions had higher p95 RTT than the corresponding default stronger-signal load conditions. They constitute labeled application traffic rather than evidence of QoS enforcement.

Each client in the full 1–100-client sweep issued 1000 requests at a nominal five requests per second (Table 7); the aggregate offered rate therefore increased from 5 to 500 requests per second. Rounding response availability to two decimal places maps fewer than 0.005% missing responses to 100.00%.

Under the stronger signal condition, response availability remained near 100% as client count increased, while p95 RTT rose most visibly for TCP at 100 clients. Under weak signal, degradation began at lower client counts and differed by protocol. MQTT p95 RTT increased from 91.8 ms at one client to 5.1 s at 100 clients. TCP retained 100% response availability but reached 2.9 s p95 RTT at 100 clients. UDP kept lower p95 RTT at high concurrency, while response availability fell from 99.5% to 90.8%. The sweep therefore

**Table 5: Attempt-weighted 5G results by workload group. RTT is sender-side request/response RTT in milliseconds; failed attempts count as unavailable and as misses at all three evaluation budgets.**

| Workload group               | Protocol and data source | Attempts | Resp. avail. | p50 RTT | p95 RTT | p99 RTT | Miss 100 ms | Miss 500 ms | Miss 1000 ms |
|------------------------------|--------------------------|----------|--------------|---------|---------|---------|-------------|-------------|--------------|
| SPaT/state-mirror probe      | 5G sender logs           | 2000     | 100.00%      | 58.7    | 130.3   | 140.5   | 46.60%      | 0.00%       | 0.00%        |
| Compact service $\leq 4$ KiB | TCP/MQTT 5G logs         | 60000    | 100.00%      | 119.7   | 413.7   | 791.7   | 62.24%      | 3.71%       | 0.55%        |
| Detector-output replay       | TCP/MQTT/UDP 5G logs     | 20100    | 99.54%       | 108.0   | 245.9   | 349.7   | 57.14%      | 0.68%       | 0.46%        |
| Map/perception 128–512 KiB   | TCP/MQTT 5G logs         | 18000    | 100.00%      | 227.2   | 668.0   | 946.0   | 99.01%      | 11.64%      | 0.73%        |
| Bulk $\geq 1$ MiB            | TCP/MQTT 5G logs         | 14000    | 100.00%      | 1146.1  | 3122.3  | 3579.1  | 100.00%     | 100.00%     | 62.61%       |

**Table 6: CAV application objects, replayed representations, and corresponding 5G workload groups.**

| CAV object                         | Source dataset/workload                            | Replayed representation                                                                                     | Source size range and percentiles                   | Measured workload group            | Budgets applied   | Application role and scope                                                           |
|------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------|-------------------|--------------------------------------------------------------------------------------|
| Periodic update or compact request | SPaT/state-mirror and compact IPI workloads        | J2735 payload or service envelope with fixed payload                                                        | 0–4 KiB fixed sizes                                 | SPaT/state mirror; compact service | 100, 500, 1000 ms | Control path with freshness or request correlation at compact sizes.                 |
| Detector object list               | V2X-Radar detector benchmark                       | Estimated result: 256-byte header plus 96 bytes per object; replayed percentiles and 60 KiB upper condition | 12.9–24.4 KiB; p50 19.2, p95 22.3, p99 23.4 KiB     | Detector-output replay             | 100, 500, 1000 ms | Post-inference structured detector output, not raw sensor data.                      |
| Camera image                       | OpenDAIR-V2X, TruckV2X, and V2X-Radar images       | Fixed 128, 256, and 512 KiB transfers                                                                       | 25.5–666.9 KiB; p50 334.1, p95 539.8, p99 648.9 KiB | Map/perception 128–512 KiB         | 100, 500, 1000 ms | A fixed transfer can represent all or part of an encoded image, depending on size.   |
| Point cloud                        | OpenDAIR-V2X, TruckV2X, and V2X-Radar point clouds | Fixed 1 and 2 MiB transfers                                                                                 | 15.0 KiB–3.85 MiB; p50 937.8 KiB, p95/p99 3.85 MiB  | Bulk $\geq 1$ MiB                  | 100, 500, 1000 ms | One transfer size or chunk; does not establish full-object completion time.          |
| HD-map object                      | OpenDAIR-V2X V2X-Seq-TFD map JSON                  | Complete size recorded; fixed 1 and 2 MiB transfers                                                         | 22.3–44.9 MiB; p50 35.8, p95 44.0, p99 44.9 MiB     | Bulk $\geq 1$ MiB                  | 100, 500, 1000 ms | Transfer chunk for a map object that requires multiple chunks at the measured sizes. |

**Table 7: Multiclient 1 KiB request/response p95 RTT (ms) and response availability.**

| Signal condition | Clients | MQTT p95 / avail. | TCP p95 / avail. | UDP p95 / avail. | Observed behavior                                        |
|------------------|---------|-------------------|------------------|------------------|----------------------------------------------------------|
| Stronger         | 1       | 40.6 / 100.00%    | 143.8 / 100.00%  | 40.5 / 100.00%   | single-client operation                                  |
| Stronger         | 2       | 42.3 / 100.00%    | 145.7 / 100.00%  | 41.5 / 100.00%   | little change from one client                            |
| Stronger         | 5       | 45.5 / 100.00%    | 145.9 / 100.00%  | 41.8 / 100.00%   | p95 remains below 150 ms                                 |
| Stronger         | 10      | 47.9 / 100.00%    | 143.9 / 100.00%  | 41.9 / 100.00%   | no missing response in this condition                    |
| Stronger         | 20      | 53.8 / 100.00%    | 145.8 / 100.00%  | 43.8 / 100.00%   | all or all but one request returns                       |
| Stronger         | 50      | 54.1 / 100.00%    | 166.1 / 100.00%  | 54.0 / 99.97%    | TCP tail begins to increase                              |
| Stronger         | 100     | 81.9 / 100.00%    | 264.1 / 100.00%  | 113.4 / 99.97%   | tail RTT increases at 500 requests/s                     |
| Weak             | 1       | 91.8 / 100.00%    | 215.7 / 100.00%  | 82.0 / 99.50%    | higher tails at one client                               |
| Weak             | 2       | 89.9 / 100.00%    | 222.6 / 100.00%  | 88.8 / 99.90%    | response availability remains above 99.9%                |
| Weak             | 5       | 112.1 / 100.00%   | 219.6 / 100.00%  | 93.0 / 99.44%    | MQTT p95 exceeds 100 ms                                  |
| Weak             | 10      | 130.0 / 100.00%   | 259.8 / 100.00%  | 104.4 / 99.65%   | all protocol conditions exceed stronger-signal tails     |
| Weak             | 20      | 175.8 / 100.00%   | 288.7 / 100.00%  | 129.4 / 99.34%   | tails increase before large response loss                |
| Weak             | 50      | 1117.7 / 100.00%  | 769.9 / 100.00%  | 169.9 / 96.73%   | MQTT/TCP tails exceed 0.7 s; UDP loses responses         |
| Weak             | 100     | 5136.0 / 99.50%   | 2908.4 / 100.00% | 172.5 / 90.81%   | MQTT/TCP tails are seconds-scale; UDP availability falls |

exposes a tradeoff between delayed responses and missing responses rather than a single client-count limit.

Receiver and broker restarts produce a separate outcome. Across the four restart conditions, 95.3–96.9% of attempts received responses. Median RTT among completed requests changed less than response availability because the requests sent during interruption timed out. These observations are timeout and restart outcomes, not explicit application rejection messages. An application must distinguish them because retry, path switching, and rejection handling are different actions.

The four restart conditions also differ in magnitude. The MQTT broker, MQTT receiver, TCP receiver, and UDP receiver restarts returned 955/1000, 991/1000, 953/1000, and

969/1000 responses, respectively. Their p95 RTTs among completed responses were 47.9, 49.9, 161.5, and 44.1 ms. The high response RTT percentiles therefore do not account for the primary restart cost; that cost appears in the missing-response count.

## 6.4 Dataset-Derived Workload Results

The workload manifest contains 22,431 staged OpenDAIR-V2X, TruckV2X, and V2X-Radar artifacts. Their sizes range from compact annotations to images of hundreds of kilobytes, approximately megabyte-scale point clouds, and HD-map JSON files of tens of megabytes (Table 6).

The V2X-Radar detector benchmark processed 922 samples without failure. The estimated serialized result sizes were 13,216–25,024 B, with p50, p95, and p99 of 19,648, 22,816, and 23,968 B. These values and a 60 KiB upper condition were replayed over TCP, MQTT, and UDP. Detector execution itself had median and p95 times of 3570.3 and 4214.4 ms. Because computation is excluded from network RTT, the replay characterizes transfer after inference. It does not show that a request-triggered sensing, inference, and response pipeline meets the 100, 500, or 1000 ms budgets; if inference is on that critical path, its measured time dominates.

Image, point-cloud, and map results have a different scope. Their source-object distributions select fixed transfer sizes, but a 1 or 2 MiB chunk does not measure completion of a 22–45 MiB map. Full-object delivery would also include segmentation, repeated transfers, reassembly, and any retransmission. The measured chunk RTT is therefore evidence about the cost of the selected transfer size, not the latency of a complete application object.

## 6.5 Measurement Scope

The measurements isolate a direct OBU–RSU exchange and a vehicle–edge exchange. V2X signal strength is normalized from the mobile runs, and the stationary payload sweeps measure delivery at the mapped signal conditions. The 5G RTT includes the vehicle modem, Airspan radio, Cisco core path, network stack, and edge responder, but excludes cloud backhaul and CAV computation. The protocol results include the implemented broker, fragmentation, and timeout behavior; the packet captures do not establish network-enforced 5QI treatment.

The 100, 500, and 1000 ms budgets are comparison thresholds rather than deadlines assigned to named driving functions. Detector replay begins after inference, and fixed large transfers do not measure multi-chunk completion. Complete application latency must include these omitted costs.

## 6.6 Summary and Discussion

Three insights follow from the combined communication measurements and workload mapping.

1) Current V2X technologies cannot support complex CAV applications. Signal strength and packet size jointly limit delivery: as normalized signal strength decreased, larger packets failed first, and even 1 KiB packets required a strong signal. Complex CAV objects begin at 13,216 B in the measured workloads and require multiple successful packet deliveries, which V2X cannot provide reliably across changing signal conditions.

2) 5G networks can support some complex CAV applications, but the data size is strictly limited by the time budget and operating condition. The compact group missed the

500 ms budget in 3.71% of attempts, whereas every transfer of at least 1 MiB missed it. Load, weak signal, concurrency, and service restarts also increased tail RTT, reduced response availability, or both.

3) Supporting collaborative CAV applications requires new networking or communication methods beyond using today's V2X and 5G network paths as-is. V2X provides direct communication for compact vehicular messages, while 5G provides addressed communication and larger transfers. Neither path alone satisfies both compact-message timeliness and request-specific delivery of larger collaborative data across the measured conditions. Candidate methods include path selection, payload reduction, caching, segmentation with completion semantics, and transfer scheduling.

## 7 Challenges and Open Questions

**OQ1: Interoperable contracts.** What cross-vendor interface should standardize freshness, request deadlines, correlation, payload, status, and failure semantics across vehicles, infrastructure, edge services, and networks?

**OQ2: Adaptive path selection.** How should an application choose or combine V2X and 5G as signal strength, payload, deadline, and recent availability change?

**OQ3: Payload adaptation.** How should a CAV system summarize, prioritize, cache, segment, or partially deliver detector, image, point-cloud, and map data without removing information required by the driving decision?

**OQ4: Failure semantics.** What evidence should distinguish rejection, expiration, timeout, packet loss, and service restart so an application can select retry, fallback, or degraded operation?

**OQ5: Evaluation methodology.** What application-aware benchmark should jointly report latency, response availability, signal and load conditions, service interruption, and complete-object delivery [17]?

## 8 Conclusion

IPI makes independent updates and correlated CAV services comparable through typed identity, timing, payload, status, and failure fields. Even 1 KiB V2X delivery required strong signal, while the smallest detector output was 13,216 B. The 5G path carried larger exchanges, but compact transfers missed the 500 ms budget in 3.71% of attempts and every transfer of at least 1 MiB missed it.

These results yield three insights: 1) current V2X technologies cannot support complex CAV applications; 2) 5G supports only the application data that fits its time budget; and 3) collaborative CAV applications require new networking or communication methods beyond either path as-is. IPI supplies fields for path selection, payload adaptation, and transfer scheduling under these constraints.

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